

Electrical Power System - Report

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1 Abstract

Munir-3 is a 1U CubeSat built to test a new antenna design where the antenna is integrated into the satellite frame itself. This report covers the design and pre-flight testing of its electrical power system (EPS), which had to deliver four separate voltage rails — 3.3V, 5V, 10V, and 12V — all from a single 3.6V battery pack. The 3.3V rail came from a linear LDO regulator, while the other three used switching boost converters. The board also carried the STM32 microcontroller acting as the onboard computer, along with an IMU and temperature sensor, and connected to the rest of the satellite through a backplane PCB.

A couple of things didn't work as expected. The original 5V regulator started shutting down once the payload and comms subsystems were pulling current together, so it got swapped out for one rated higher. The 12V regulator was also damaged early on due to a pin wired to the wrong voltage, which was fixed by rerouting it to the battery instead. The backplane needed a few tweaks too, mainly correcting some mirrored connectors and moving the battery connections to through-holes for easier soldering.

On the ground, the EPS did what it was supposed to: each rail hit its target voltage, the STM32 ran fine, and the satellite managed to collect and send data to a dummy ground station during isolated testing. It also survived vibration testing before launch. The satellite launched as part of NASA's GHOST mission, but no data ever came back afterward, so whether the EPS actually performed in orbit is still unknown. Most of what got fixed along the way was caught through testing rather than design review, which says something about the value of testing subsystems together under real, combined loads rather than trusting datasheets alone.

2 Introduction

CubeSats place strict constraints on subsystem design because of their small size and limited available power. For this reason, the electrical power system (EPS) is a central part of the satellite, since it must provide stable power to the onboard electronics and support reliable operation of the system as a whole. Munir-3 is a 1U CubeSat developed to test a new antenna architecture in which the antenna is integrated into the satellite frame. In this project, the main focus of the EPS subsystem was to design the main EPS board. The board was designed to provide 3.3V for most functions on the PCB, 5V for external subsystems, and 10V and 12V for the payload. In addition, the board included an STM32 microcontroller acting as the onboard computer, together with an Inertial Measurement Unit (IMU) and a temperature sensor. A backplane was also designed to connect the main EPS board to the other subsystems and carry both power and buses. In this way, the EPS was not only responsible for power distribution, but also formed an important part of the overall system integration. Before launch, the satellite was tested in isolation and as an integrated system. These tests showed that the board could support data collection, onboard processing, and transmission to a dummy ground station. Environmental testing, including vibration tests at Kongsberg and through NASA as part of the launch campaign, was also completed before flight. The satellite was launched as part of NASA's GHOST mission, but no data were received after launch. Since the cause of this is outside the scope of the present work, the report focuses on the EPS design and its demonstrated performance during pre-flight testing.

3 Theory

This section presents the theoretical background needed to understand the design choices described in the Methods section, covering the basic operating principles of linear and switching voltage regulators, and the role of the electrical power system within a CubeSat.

The electrical power system (EPS) is one of the core subsystems of a CubeSat, responsible for distributing power from the battery to all other subsystems on board. Most onboard electronics, sensors, and communication modules need different supply voltages to work properly, so the EPS has to convert the battery's nominal voltage into several regulated output rails. Beyond just distributing power, the EPS often takes on other jobs too, such as running an onboard microcontroller for system control and monitoring, or hosting sensors for telemetry data like temperature or orientation. Because the EPS connects to nearly every other subsystem, both electrically and often physically through a backplane, how well it's designed has a direct effect on how the whole satellite performs. Meeting these different voltage requirements calls for specific types of regulators, which are explained below.

In the Munir-3 EPS module, four regulated voltage rails were needed: 3.3V, 5V, 10V, and 12V. The 3.3V rail powered most of the components on the EPS board itself, including the STM32 microcontroller, while the 5V rail supplied external subsystems such as the communications module. The 10V and 12V rails were dedicated to the payload. All four rails were derived from a single battery pack with a nominal voltage of 3.6V. Since the battery voltage doesn't match any of these required outputs, and it drifts as the battery charges and discharges anyway, voltage regulators are needed to turn it into something stable and fixed. Depending on whether the target voltage is below or above the battery voltage, a different type of regulator is used: linear regulators for stepping down, and switching boost converters for stepping up.

3.1 Low-dropout regulator (LDO)

An LDO is a linear voltage regulator that produces a stable output voltage lower than its input. It works by placing a variable-resistance element, usually a transistor, in series between input and output, and constantly adjusting that resistance to keep the output steady even as the input voltage or load current changes. The extra voltage, the difference between input and output, just gets dissipated as heat. This makes LDOs simple and quiet from an electrical noise standpoint, but they become less efficient as the gap between input and output grows, since all of that gap turns into wasted heat rather than usable power. The "low-dropout" part of the name refers to how little headroom the regulator needs between input and output to keep working, compared to older linear regulators that needed a much bigger margin.

3.2 Boost converter

A boost converter, also called a step-up converter, is a switching regulator that produces a higher output voltage than its input. Instead of burning off extra voltage as heat the way a linear regulator does, it moves energy around using an inductor, a switch, and a diode (or a synchronous

switch), storing and releasing that energy in quick bursts at a fixed switching frequency. While the switch is on, current builds up in the inductor and stores energy in its magnetic field. When the switch turns off, the inductor releases that stored energy, and because it resists any sudden change in current, it pushes the voltage higher to keep that current flowing. This energy then flows through the diode to the output, where a capacitor smooths it out into a stable DC voltage. By repeating this switching cycle very fast and controlling the ratio of on-time to off-time, called the duty cycle, the converter can be tuned to hit whatever output voltage is needed above the input. Since energy is transferred rather than burned off, boost converters can be a lot more efficient than linear regulators when stepping up voltage.

3.3 Feedback voltage divider

To keep its output steady, a boost converter has a feedback pin (FB) that watches the output and adjusts the duty cycle to correct it. The catch is that the FB pin expects a much lower voltage than what's actually coming out of the regulator, usually just an internal reference of around 1 V. To handle this, a resistive voltage divider is connected across the output, with the midpoint tied to the FB pin, scaling the output down to something the regulator can compare against its reference. By picking the two resistor values, you can set the output to whatever voltage you need: a bigger ratio between the resistors pushes the output higher, since more of it has to get divided away to land back on that same fixed reference.

This relationship comes from the standard voltage divider equation, $V_{OUT} = V_{FB} \times (1 + R1/R2)$, where V_{FB} is the regulator's fixed internal reference and $R1$, $R2$ are the two resistors making up the divider. In practice, $R2$ is usually picked first, low enough that the current through the divider stays small compared to the load, but not so low that it wastes power, and $R1$ is then worked out to hit the target voltage. Since only the ratio between the resistors matters and not their absolute values, the same output voltage can be reached with plenty of different resistor combinations, so component tolerance and what's actually available in standard resistor values end up playing a role too.

4 Methods

The Munir-3 needed 3.3V, 5V, 10V and 12V outputs for its different subsystems. Each voltage group has its own voltage regulators and converters and its own circuit on the EPS board.

The 3.3V LDO (Low Drop out) regulator circuit that uses the MIC29300-3.3WU which is a high-current low dropout regulator with high accuracy. The MIC29300 has a current capability of 3 amps. The regulator is applicable in low current applications as well, especially when small dropout voltages and ground current values are important (Microchip Technology Inc., 2019).

The 5V rail initially used the MIC2876-5.0YMT-TR, a synchronous boost converter, to step up the battery's nominal voltage of 3.6V to a regulated 5V output, rated for up to 1.2A (Microchip Technology Inc., 2016). This regulator was later replaced with the TPS61232DRCR, dimensioned to supply up to 2.1A, to ensure sufficient headroom for the combined current draw of the payload and communications subsystems during nominal operation (Texas Instruments Incorporated, 2014). The 10V and 12V rails both used the ADP1610ARMZ-R7, a PWM current-mode DC-DC boost converter capable of providing output voltages up to 12V from an input range of 2.5V to 5.5V, with up to 92% efficiency (Analog Devices, Inc., 2021). Both circuits stepped up from the battery nominal voltage of 3.6V. The output voltage was set by a resistive voltage divider connected to the feedback pin (FB), using the relation:

$$V_{OUT} = V_{FB} \times \left(1 + \frac{R_1}{R_2}\right) \quad (1)$$

where $V_{FB} = 1.23V$ was the internal feedback reference voltage of the ADP1610.

For the 10V rail, $R_1=71.5k\Omega$ and $R_2=10k\Omega$ were selected, yielding a designed output voltage of:

$$V_{OUT} = 1.23 \times \left(1 + \frac{71500}{10000}\right) = 10.02V \quad (2)$$

For the 12V rail, $R_1=88.7k\Omega$ and $R_2=10k\Omega$ were selected, yielding:

$$V_{OUT} = 1.23 \times \left(1 + \frac{88700}{10000}\right) = 12.14V \quad (3)$$

Both circuits used a $1\mu H$ inductor in the switching stage and a Schottky diode as the output rectifier. Output filtering consists of a $22nF$ and $10\mu F$ capacitor in parallel at the input stage, and a $100nF$ and $10\mu F$ capacitor in parallel at the output. A $150pF$ capacitor was connected to the SS pin on each circuit to control the soft-start ramp, limiting inrush current at startup. Both rails supply the payload subsystem and have a maximum output current of 300 mA. The 3.3V rail uses the MIC29300-3.3WU, a fixed-output low dropout (LDO) linear regulator with a maximum output current of 3A. Unlike the boost converters used for the other rails, the LDO regulated down from the battery voltage by dissipating the excess voltage as heat. Input and output decoupling was provided by a $10\mu F$ and $100nF$ capacitor on each side. The 3.3V rail powers the majority of the components on the EPS board itself. Each voltage rail included a dedicated status LED connected through a 470Ω current-limiting resistor, providing a visual

indication that the rail was active.

4.1 Experimental setup

The EPS board was mounted to the satellite frame using stand-off screws on each corner of the board with an approximate area of $68 \times 70 \text{ mm}^2$ and a board thickness of 1.46 mm. A backplane PCB was connected to the EPS board using a 30-position 2.54 mm pitch pin header, highlighted by the red square in fig. 2, interfacing the EPS board with the other subsystems of the satellite. The backplane PCB is mostly passive, but houses two ISA150233C03LMDSXTMA1 dual N+P channel MOSFETs responsible for power switching (Infineon Technologies AG, 2024). The first controls the main battery connection to the EPS board, and the second controls the burn wire circuit used for antenna deployment. Connections to the battery are made through through-hole pads rather than surface pads, allowing wires to be soldered securely for the duration of the mission. These connections use standard 26 AWG cabling, rated for 2A, chosen for its simplicity and its demonstrated ability to withstand vibration testing. The burn wire circuit, which requires a higher current, uses a minimum of four such cables to safely handle the expected load. Prior to powering the board, continuity checks were performed using a multimeter to verify there were no short circuits between power rails and ground. The board was then powered using a bench power supply set to 3.6 V to simulate the nominal battery voltage. Each voltage rail was subsequently measured using a multimeter to verify the output voltage of the 3.3V, 5V, 10V and 12V regulators.

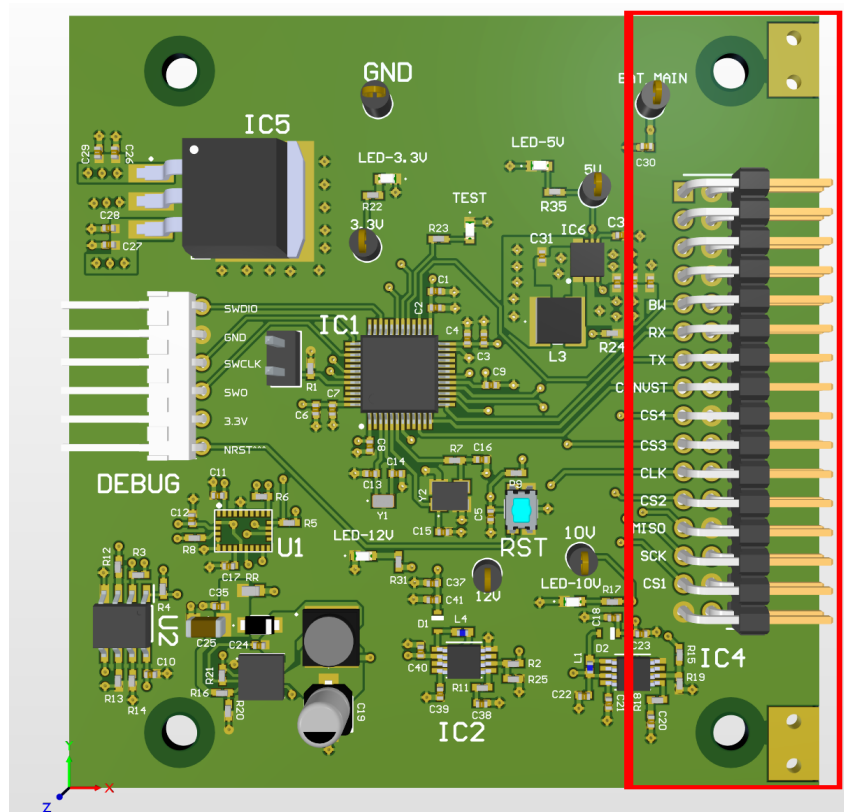


Figure 1: EPS Main PCB Render

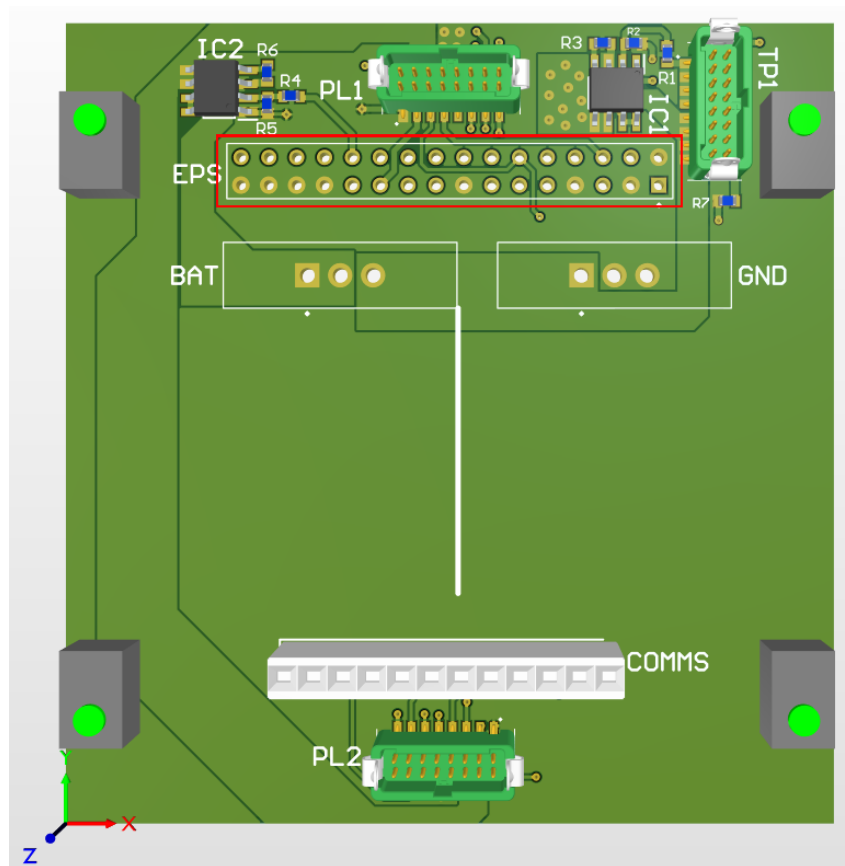


Figure 2: Backplane PCB Render

5 Results

The results presented here are based on ground testing and system verification before launch. Since no data were received in orbit, the performance of the EPS can only be evaluated from pre-flight results.

5.1 EPS performance

During testing, the main EPS board provided the intended voltage levels required by the satellite. The 3.3V rail powered the main board functions, while 5V, 10V, and 12V were supplied as required by the connected subsystems. This showed that the board was able to perform its main task as the central power-distribution unit of the satellite. The STM32 also operated as intended during testing, and the board functioned as a complete integrated unit together with the sensors and backplane connections. From a design perspective, this was an important result, since it showed that the PCB could support both power distribution and onboard control within the same compact system.

During testing of the initial 5V regulator (MIC2876-5.0YMT-TR), the rail entered shutdown mode above roughly 0.9A of load current, despite a 1.2A rating stated in the datasheet. Since the payload and comms subsystems combined were expected to draw more current than this during nominal operation, with comms alone requiring 0.8A, the regulator was replaced with the TPS61232DRCR, dimensioned to supply 2.1A. This replacement was implemented following the datasheet's reference layout and was subsequently tested.

The 12V rail also required a design correction. In the initial board revision, the boost regulator's BOOT, VDD, and EN pins were connected to the regulator's output, but these pins could not tolerate 12V, which led to IC damage and a short across the PCB. This was fixed by cutting the affected traces and reconnecting these pins to the battery supply instead, which resolved the issue in later board revisions.

5.2 Backplane

The backplane's main function is to connect all the different subsystem PCBs together, including the battery, into a single integrated structure. Testing and review of the backplane also turned up a few improvements over the previous iteration. The payload and TP1 connectors, which had been mirrored the wrong way in the earlier design, were checked and confirmed to now be correctly oriented. The battery connection pads were changed from surface pads to through-holes, which made it much easier to solder the wires reliably and should hold up better for the rest of the mission. Standard 26 AWG cabling, rated for 2A, was used for these connections instead of specialized connectors, since this approach had already survived vibration testing in the previous iteration. For the burn wire circuit, which draws more current, at least four cables were used to safely handle the expected load. The physical dimensions of the backplane PCB, including the mounting hole positions, were left unchanged from the previous iteration, since these had already been measured and verified to a high level of precision. A

backup communication subsystem from the previous iteration was also removed, a crystal oscillator and transistor circuit meant to let comms talk directly to the payload if the STM32 ever became unreachable, since it was no longer needed and its components would have physically collided with the comms board anyway.

5.3 System-level verification

In isolated testing, the satellite was able to collect, process, and send data successfully to a dummy ground station. This indicates that the EPS design worked as intended under controlled conditions and that the main board supported stable operation of the onboard system. Environmental testing was also completed before launch, including vibration testing. This provided additional confidence that the hardware was mechanically robust enough for launch conditions.

5.4 Limitations of the results

Although the pre-flight results were positive, no data was received after launch. This means that the EPS could not be fully verified under real mission conditions. The results are therefore somewhat limited: they show that the design worked on the ground, but they do not allow firm conclusions about in-orbit performance. Overall, the results suggest that the EPS design was successful at board and system level during testing. At the same time, the lack of flight data means that the final operational performance of the system remains unresolved.

6 Discussion

This section discusses the results presented above, reflecting on the design issues encountered during development, what they reveal about the EPS design and verification process, and the limitations of evaluating the system without in-orbit data.

6.1 Regulator problems

Both the 5V and 12V regulator issues stemmed from decisions made early in the design process that were not fully verified against the datasheets before the board was built. In the case of the 5V rail, the original regulator's datasheet current rating did not hold up under the actual combined load of the payload and comms subsystems, which only became apparent once the board was tested with these subsystems connected rather than in isolation. Similarly, the 12V regulator failure resulted from connecting the BOOT, VDD, and EN pins directly to the output voltage, a configuration that exceeded the maximum voltage these pins could tolerate. This suggests that a more thorough pin-by-pin review against the datasheet's absolute maximum ratings during the design phase could have caught this issue before it caused physical damage to the board. Both issues were resolved through iterative testing and design revisions rather than being identified during the initial design review, highlighting the value of testing subsystems together under realistic load conditions rather than relying solely on datasheet specifications in isolation.

6.2 Backplane redesigning

Similar to the regulator issues, most of the backplane changes came out of lessons learned during testing rather than anything caught in the initial design review. The payload and TP1 connectors being mirrored the wrong way, for example, wasn't noticed until the physical board was reviewed, which suggests these footprints and orientations could have been checked more carefully against their mating components earlier on.

Other changes were more about practicality than fixing something broken. Switching from surface pads to through-holes for the battery connections was really about making the soldering job easier, not about a functional failure, and removing the backup communication subsystem came down to it no longer being needed and getting in the way of the comms board, rather than any fault with it. Put together, these changes follow the same pattern as the regulator problems: most of what got fixed was found by actually building and testing the thing, not by catching it on paper beforehand.

6.3 Limitations of missing flight data

A key limitation of this work is that everything here rests on pre-flight, ground-based testing. No data came back after launch, so there's no real way to confirm the board did what it was supposed to once it was actually in orbit.

Some of that gap is just unavoidable. Things like vacuum, radiation, and thermal cycling weren't directly replicated during testing beyond the vibration tests done before launch, so we simply don't know how they affected the EPS. Time was tight too: fixes like the 5V regulator swap were only tested within a short timeframe before launch, which didn't leave much room to see how reliable they'd hold up over the mission.

That said, there's some reassurance from outside the EPS itself. The plasma-sensing probes were tested and confirmed working in a simulated plasma tank, and later the whole satellite went through the same tank testing, since this environment is meant to mimic low Earth orbit fairly closely. This result was reaffirmed after launch, when further lab-based plasma testing again confirmed correct operation of the sensing hardware. It's a good sign that the sensing hardware, and the power feeding it, should behave the same way once actually in orbit.

Ground testing gives a reasonable level of confidence, but it's not a stand-in for the real thing. What the EPS actually did in orbit is still an open question.

6.4 Future Work

Going forward, figuring out why no data came back after launch is the top priority, since that's still unresolved and wasn't something this report could dig into. Looking into the communications chain, from the STM32 all the way to the ground station link, would help pin down whether the problem was with the EPS, another subsystem, or the link itself.

On the EPS side, it'd be worth running the replacement 5V regulator through longer testing under realistic combined loads, since there wasn't much time to do that before launch. The 10V and 12V rails could use the same treatment too. A quick bench measurement is fine for a first check, but sustained load testing would help rule out any issues that just haven't shown up yet.

It'd also help to build a more systematic design review process into future iterations, something like a checklist that cross-checks every pin connection against the datasheet, so problems like the 12V regulator's pin setup get caught before the board is built instead of after.

7 Conclusion

This report covered the design and pre-flight testing of the electrical power system (EPS) for the Munir-3 CubeSat. The board was built to provide 3.3V, 5V, 10V, and 12V rails to the satellite's subsystems, alongside the STM32 microcontroller, IMU, and temperature sensor, with a backplane tying it all together with the rest of the satellite.

Testing showed the EPS did what it was supposed to. Each rail put out its intended voltage, the STM32 worked fine, and during isolated testing the satellite managed to collect, process, and send data to a dummy ground station without issue. It also made it through vibration testing before launch.

Along the way, a few things went wrong and got fixed, though mostly through trial and error rather than being caught up front. The original 5V regulator kept shutting down once the payload and comms were pulling current together, so it got swapped for one that could handle more. The 12V regulator ended up damaged from a pin wired the wrong way, which was fixed by rerouting those pins to the battery instead. The backplane got a few tweaks too, connectors that had been mirrored the wrong way were corrected, and the battery connections moved to through-holes for easier, more reliable soldering.

No data ever came back after the satellite launched as part of NASA's GHOST mission, so what actually happened to the EPS in orbit is still unknown. There's some reassurance from elsewhere though: the plasma probes and later the whole satellite performed well in simulated plasma tank testing, and this was reaffirmed by further lab-based plasma testing after launch, which is a decent sign the hardware should hold up in orbit too. All in all, the pre-flight results show the EPS worked the way it was meant to, and the problems that came up along the way are a good reminder of why testing things under real, combined conditions matters more than checking datasheets alone.

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